



Regulation of entry, labor market institutions and the informal sector[☆]

Gabriel Ulyssea^{*}

The University of Chicago, Department of Economics, 1126 E. 59th St., Chicago, IL 60637, United States
IPEA

ARTICLE INFO

Article history:

Received 21 June 2006

Received in revised form 3 July 2009

Accepted 6 July 2009

JEL classification:

J41

O17

Keywords:

Informal sector

Institutions

Labor market

ABSTRACT

This paper develops a two-sector matching model that incorporates the main features of Latin American labor markets. It has an innovation in its matching structure that makes it more consistent with some key stylized facts of the informal sector in these countries. The model is numerically solved using Brazilian data and several policy simulations are performed. Reducing formal sector's entry cost significantly reduces the size of the informal sector and improves overall labor market performance. Increasing enforcement significantly reduces informality but has strong adverse effects on unemployment and welfare. Thus, the results indicate that the tradeoff between lower informal employment and higher unemployment rates is not present when one looks at policies that aim at reducing the costs of being formal, as opposed to policies that simply increase the costs of being informal.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

A common feature of almost every Latin American country is the existence of extremely restrictive labor market institutions and strict regulation of entry. As a consequence, for both firms and workers the decision of being formal turns out to be extremely costly, which significantly increases the attractiveness of informal activities.

From the firms' perspective, the costs of being formal can be separated in two: (i) the costs of entering the formal sector, which come from procedures, fees and bureaucracy requirements to start a formal business; and (ii) the costs of staying in the formal sector, which can be further divided into taxes, regulation and bureaucratic requirements. Being informal also entails costs that are directly or indirectly related to labor market institutions: (i) the official penalties applied when the firm is caught by the government; and (ii) restricted access to public goods, specially the law system and police. Turning to the workers side, the main costs of being formal come from the taxes associated to formal labor contracts, notably income taxes and mandated social security contributions. Being informal, however, implies that the worker does not have access to the benefits predicted by law, is not protected by job security legislation and faces higher turnover rates.

The objective of this paper is to analyze the impact of different labor market institutions and regulation of entry on the size of the informal sector and overall labor market performance in an integrated framework. For that, I extend the two sector continuous-time matching model proposed by [Acemoglu \(2001\)](#) in order to incorporate the main institutional aspects of most Latin American labor markets. Additionally, I modify its matching structure so that it combines separated markets and undirected search, which makes it more coherent with some key stylized facts of the informal sector in developing countries. The analysis is focused on steady state equilibria and, therefore, it regards the long term effects of institutions.

The model is numerically solved using Brazilian data and used to quantitatively analyze the impact of several policies. Brazil constitutes an interesting case study, as it has a large informal sector nearly 50% of its labor force and very restrictive labor and entry regulations (it is ranked among the most restrictive in Latin America). Moreover, the informal sector has been steadily increasing over the past decade and this growth started just after a major revision of the national Constitution in 1988, which significantly increased overall labor costs.

The simulation results indicate that reducing payroll taxes and increasing unemployment benefits are not effective policies to reduce informality and improve labor market performance. In contrast, lowering the costs of entry into the formal sector substantially improves employment composition: making formal sector's entry costs similar to informal sector's would increase formal employment by nearly 31% and reduce unemployment by 36%. In fact, reducing entry costs improves all labor market indicators and increases welfare. Importantly, increases in unemployment resulting from raising formal

[☆] I am deeply indebted to Ricardo Paes de Barros and José Márcio Camargo for their guidance, comments and suggestions. I would also like to thank Carlos Tomei, Gustavo Gonzaga, Mauricio Cortez, Carlos H. Corseuil, Sarah J. Cattán and two anonymous referees for their comments. All remaining errors are mine.

^{*} The University of Chicago, Department of Economics, 1126 E. 59th St., Chicago, IL 60637, United States. Tel.: +1 773 895 6245; fax: +1 773 702 8490.

E-mail address: ulysssea@uchicago.edu.

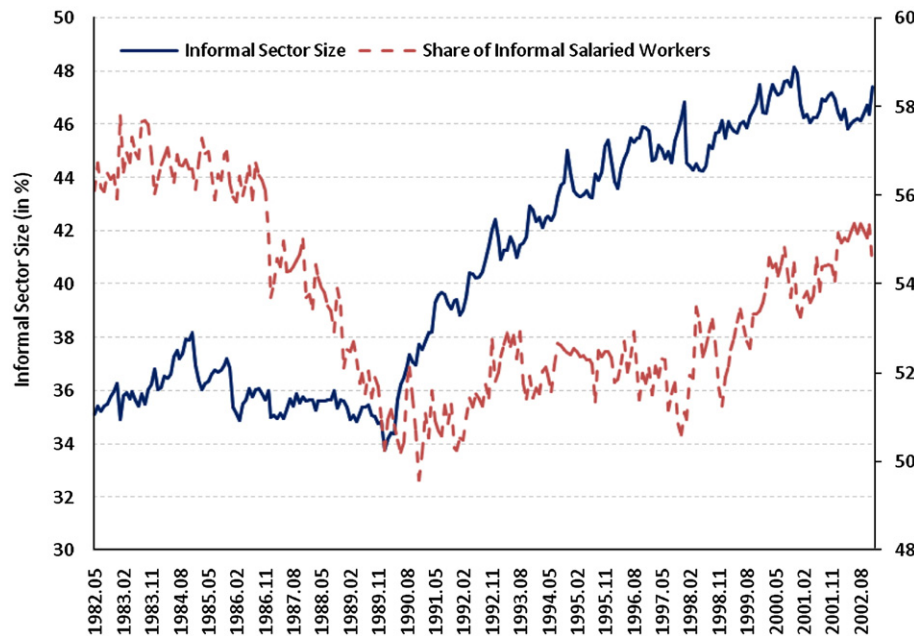


Fig. 1. Informal sector evolution in Brazil. Source: Monthly Employment Survey (PME). Secondary y-axis: Informal salaried employees as a share of total informal employment (in %).

entry costs are also accompanied by increases in the wage gap.¹ Thus, the model's results are in line with the empirical relationship between the controlled wage gap and unemployment documented since the early 1990s (see Section 2). Considering the substantial increases in the fixed costs of hiring formally observed in the 1988–1998 period (see Gonzaga, 2003), these results offer a rationale for the aggregate behavior of the Brazilian labor market during the 1990s and early 2000s.

The results also indicate that increasing the enforcement of current labor regulations is very effective to reduce the size of the informal sector, but it also significantly increases unemployment and leads to substantial welfare losses: increasing enforcement could cause a decrease of up to 14.3 percentage points in the size of the informal sector, but it would also imply a 6 p.p. increase in unemployment and a nearly 15% decrease in welfare. These adverse effects are a result of stronger enforcement of current inadequate labor market institutions and high entry costs in the formal sector.

These results could in principle be seen as supportive of the argument that rationalizes the so-called “shadow puzzle”: large informal sectors are widely tolerated in spite of the improvements in technologies to detect shadow activities because intensifying the degree of coercion would lead to higher unemployment (see Boeri and Garibaldi, 2005). Taken together, however, my results regarding formal sector's entry cost and enforcement level make the latter argument somewhat less appealing. The results show that the tradeoff between lower informal employment and higher unemployment rates does not exist when one looks at policies that reduce the cost of being formal, instead of repressive or punishment policies. Thus, the best option to decrease informality and improve labor market performance and welfare would be to reduce the costs of entry into the formal sector instead of intensifying punishment and auditing of informal activities.

This paper relates to the literature that focuses on the causes and consequences of the informal sector and, more specifically, to the studies that analyze the role played by institutions. A number of recent papers have analyzed different aspects of informality within

the framework of matching and search models. Boeri and Garibaldi (2005) and Albrecht et al. (2009) analyze how workers with heterogeneous productivity sort into both sectors and how some specific labor market policies may affect informal sector's composition and size, as well as their impact on unemployment level. Bosch (2006) focus on the within firm margin of informality to analyze why the job finding rate of formal jobs is strongly pro-cyclical and volatile, while in the informal sector this rate is relatively stable. Finally, Zenou (2008) develops a model with search frictions in the formal sector, whereas the informal sector is fully competitive, to assess the impact of policies aiming at reducing unemployment.

To the best of my knowledge, the model developed in this paper is the only one in the literature of formal–informal matching models to combine separated markets and undirected search. This constitutes an advantage, as it allows to analyze both sectors separately, and hence to compute separated tightness, vacancy rates and employment probabilities but without assuming that workers must direct their search to a specific sector. Thus, this approach allows workers to apply for both sectors simultaneously, which seems to be more reasonable when one analyzes labor market informality in developing countries (see, for instance, Maloney, 1999). In addition, this structure allows me not to rely on sectorial unemployment rates, which are hard to justify empirically.

Moreover, evidence for developing countries seems to indicate that perfect competition and full employment do not adequately depict informal sector's functioning. Labor market frictions in the informal sector might be less important than in the formal sector, but they seem to be too significant to be assumed away. Flows from informal employment to unemployment are quite sizable and flows out of unemployment to formal and informal employment are nearly of same magnitude, which suggests that modeling the informal sector as a frictionless buffer might not be a good approximation.² This model is in line with this evidence, as it provides a rationale for the existence of the informal sector that does not need to rely on technological differences or labor market imperfections in the formal sector vis-à-vis a perfectly competitive informal sector.

¹ Given that workers are homogeneous in this model, the empirical counterpart of the wage gap observed in the model would be the controlled (or conditional) wage gap.

² Maloney (1999) provides detailed evidence for Mexico.

Another aspect worth highlighting is that the model does not rely on workers' heterogeneity to generate wage differentials between formal and informal workers that match the data. Thus, the model provides a framework that justifies the high formal–informal wage gap between observably identical workers solely through labor market institutions. Finally, the model's structure provides a natural way to evaluate the net aggregate impact of changes in a broad array of labor market institutions on the economy's welfare and productivity. In particular, there is no need to assume *a priori* that a larger share of formal jobs is socially preferred per se; it will be the case as long as it is accompanied with a general improvement of labor market performance.

The remainder of this paper is organized as follows. Section 2 presents some stylized facts of the informal sector in Brazil and provides some institutional background. Section 3 analyzes the simulation model. Section 4 briefly presents the data and carefully discusses the quantitative results. Finally, Section 5 concludes.

2. Stylized facts

The informal sector is usually defined as all economic activities that contribute to the gross national product, but escape detection in the official GDP estimates (see, for instance, Schneider and Enste, 2000). However, this traditional definition is too broad as it includes both legal and illegal activities and it is especially inadequate when one analyzes the relationship between institutions, informal employment and labor market performance. For this reason, I use a more restrictive definition that considers as part of the informal economy only the legal but unregulated activities. To disentangle legal from illegal activities, I measure the informal sector as the proportion of unregulated workers that are engaged in legal occupations. I do so by defining the informal sector as the share of workers who do not contribute to the social security agency (this is a common definition in the literature, see for instance Boeri and Garibaldi, 2005; Bosch, 2006). This definition pools informal salaried workers (who do not have the mandatory work permit) and the self employed workers together. Although these two types of workers differ in some important aspects, they share the fundamental characteristic of having an extremely flexible and unprotected type of occupation, and hence can be analyzed together.³

The first piece of evidence worth highlighting is the sharp increase in informal employment that was observed in the last decade, as shown in Fig. 1.⁴ This trend is in line with the well documented growth of the informal sector in both developed and developing countries in the past decade (Schneider and Enste, 2000; Boeri and Garibaldi, 2005). Nonetheless, this phenomenon seems to have been much stronger in Brazil than in most countries. Interestingly, this trend becomes noticeable just after the major revision of the Brazilian Constitution in 1988, which significantly increased the cost of formal employment.⁵ This period was also characterized by other important transformations in the Brazilian economy, such as the privatization process and trade liberalization. Nonetheless, recent evidence indicates that trade liberalization had little impact on informal employment's growth, while changes in labor market regulation seemed to have played a major role (Goldberg and Pavcnik, 2003; Bosch et al.,

Table 1
Regulation costs.

Panel A: Labor costs			
	Hiring index	Firing index	Payroll taxes
Entire sample (155 countries)			
Mean	36.7	35.9	16.3
1st quartile	11.0	20.0	8.0
2nd quartile	33.0	40.0	14.7
3rd quartile	61.0	50.0	23.8
Maximum	100.0	100.0	55.0
Brazil	67.0	20.0	26.8
Latin America	40.5	29.5	15.9
OECD: high income	30.1	27.4	20.7
Panel B: Entry cost			
	Procedures ^a	Duration ^b	
Entire sample (155 countries)			
Mean	9.5	47.5	
1st quartile	7.0	24.0	
2nd quartile	9.0	38.0	
3rd quartile	12.0	56.0	
Maximum	19.0	203.0	
Brazil	17.0	152.0	
Latin America	11.4	63.0	
OECD: High income	6.5	19.5	

Source: Doing Business database, available at www.doingbusiness.org.

^a Number of procedures necessary to start a business.

^b Number of days to start a business.

2007). A second fact worth noting in Fig. 1 is that the increase in informal employment across the 1990s was accompanied by an increase in the share of informal salaried workers in total informal employment. After having dropped significantly during the 1980s, employment across the proportion of informal salaried workers in total informal employment increased substantially during the following decade (with somewhat strong oscillations during the period) and reaching the end of 2002 corresponding to 55% of total informal employment.

A second important stylized fact in the informality literature is that a heavier and stricter regulation is correlated with a larger informal sector.⁶ In Brazil, besides the costs imposed by the labor legislation, firms also face significant entry costs to the formal sector in the form taxes and bureaucracy burden, as shown in Panels A and B of Table 1, respectively.⁷ Indeed, except for the firing index, Brazil's indicators are well above the Latin American average, being located at the top quartile of the distribution. Looking only to start-up costs, the country's indicators are close to the maximum values observed in all the 155 countries for which there the data is available.

Nonetheless, these high costs and rigidities imposed by regulation do not seem to be generating severe barriers to workers' mobility between formal and informal sectors. Table 2 presents the mobility pattern of Brazilian workers from 2003 to 2005, which was estimated from the rotating panel of the Monthly Employment Survey (PME). This matrix displays the transition rates between the main employment and unemployment status in the Brazilian labor market.⁸ The rows indicate the original status and the columns the worker's destination after a 12 month period. Therefore, the diagonal's

³ Bosch et al. (2007) empirically analyze these two types of informal workers separately, while Bosch (2006) formalizes the decision of becoming self employed.

⁴ Fig. 1 uses the data from the Brazilian Monthly Employment Survey (*Pesquisa Mensal do Emprego* – PME), which is a rotating panel that has the same structure as the Current Population Survey (CPS) and it tracks the same individual for a maximum period of 16 months.

⁵ Barros and Corseuil (2001) provide a detailed description of all the labor-related modifications made in the 1988 Constitution.

⁶ Using cross-country data, Djankov et al. (2002) find that stricter entry regulation is associated to a greater relative size of the unofficial economy and larger employment in the informal sector. Similarly, Auriol and Warlters (2005) argue that a major determinant of micro-enterprises informality in developing countries is the existence of high fixed costs involved in becoming formal.

⁷ The data used to construct Table 1 was obtained from the Doing Business project (www.doingbusiness.org) of the World Bank. The methodology of the indexes used here can be found in Djankov et al. (2002) and Botero et al. (2004).

⁸ The other possible statuses (out of the labor force, employer, public employee and so forth) were excluded from Table 2 because they are not directly related to the main focus of this paper.

Table 2
Transition matrix (in %).

	Formal	Informal	Unemployed	Self employed
Formal salaried	84.9	7.2	4.7	3.2
Informal salaried	24.3	53.6	8.4	13.7
Unemployed	23.5	25.3	38.7	12.6
Self employed	7.1	13.2	4.0	75.7

Source: Monthly Employment Survey (PME).

Note: Includes 13 panels ranging from Jan/03 to Jan/05.

elements can be seen as an approximation of the probability of staying in the same status after a 12 month period or, roughly, the complement of the exit probability of any given state. As expected, the formal sector presents a significantly lower exit probability than the informal one, but the transition rate from the formal to the informal sector is much lower than in the opposite direction. This suggests that workers' mobility is not as restricted as the traditional dualistic view would predict.⁹

Turning to the analysis of wage differentials, Fig. 2 shows a clear downward trend in the gross formal–informal wage gap (without any control for observed characteristics) starting in 1992. In contrast, since 1995 the unemployment rate has shown an upward trend, with a total increase of nearly 45%. Putting these two facts together, one obtains the negative correlation of formal–informal wage gap and unemployment rate presented in Boeri and Garibaldi (2005). However, once controlling for workers observable characteristics in a simple Mincerian regression,¹⁰ a different scenario arises: from 1995 on, the behavior of the gross and controlled wage differential starts to diverge significantly (Fig. 2). In contrast, the controlled wage gap and unemployment rate start to show a very similar pattern: Instead of the usual negative correlation between the wage gap and unemployment rate, from 1995 on there is a positive correlation between unemployment and the controlled wage gap (Fig. 3).

Finally, it is worth highlighting that even after controlling for individuals' observable characteristics the wage gap remains considerably high (around 30%). Therefore, workers' observables only account for a limited fraction of the gross wage gap, which suggests that there is more than just workers' heterogeneity behind the total wage gap.¹¹

3. The model

3.1. The production side

The production side in this model assumes that two goods are produced: a consumption good and an intermediate good. The latter can be produced in the formal or in the informal sector, while the consumption good is produced by a single representative firm that uses a CES function defined over the two intermediate goods produced in the formal and informal sectors:

$$Y = (aY_F^\rho + (1-a)Y_I^\rho)^{\frac{1}{\rho}} \quad (1)$$

where Y_F and Y_I denote the aggregate output of the intermediate good in the formal and informal sectors, respectively. The parameter $0 < a < 1$

⁹ Maloney (1999) discusses the relationship between workers' mobility and the dualistic view of formal–informal labor market segmentation.

¹⁰ The control variables are the usual ones: gender, age, age square, education, education square, region and sector dummies, position in the household and a dummy if the individual lives in a metropolitan region.

¹¹ Other studies indicate that even after controlling for sample selection bias the wage differentials between formal and informal workers remain large. In fact, some authors find that OLS actually underestimates the true wage differential (see, for instance, Carneiro and Henley, 2001).

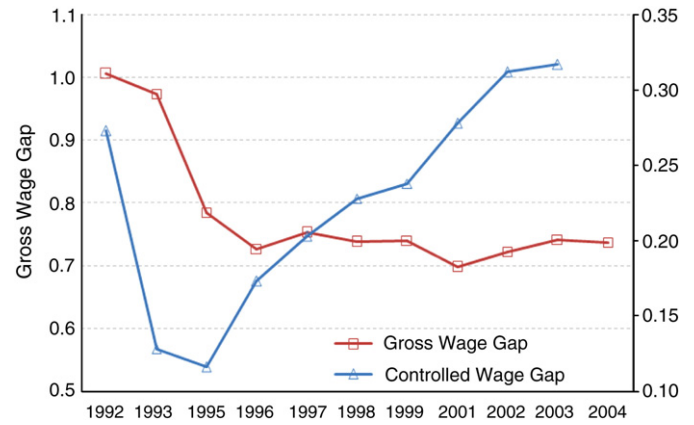


Fig. 2. Gross versus controlled formal–informal wage differential.

measures the relative importance of the aggregate formal product (Y_F) in the production of the final good (Y) and the parameter ρ determines the elasticity of substitution between Y_F and Y_I , which is given by $1/(1-\rho)$.

As I discuss in Section 3.5.3, the elasticity of substitution plays an important role in determining whether or not the model presents a unique equilibrium. More specifically, as $\rho = 1$ (and thus formal and informal intermediate goods are perfect substitutes) the model presents multiple equilibria, which is consistent with the analytical result in Acemoglu (2001). Throughout the simulation exercises (Section 4), I assume that formal and informal inputs are not perfect substitutes, which seems a more plausible assumption. As pointed out by Acemoglu (2001), this formulation captures the idea that there is some need for diversity in overall production, which seems coherent with the empirical evidence of a sizable informal sector in all Latin American countries.¹² Additionally, Bosch (2006) shows that even within firms both formal and informal labor workers are employed in production.

Throughout this paper the price of the final good is normalized to one. Both intermediate goods are sold in a competitive market and cannot be stored. Consequently, the prices are given by their respective marginal productivity:

$$p_F = aY_F^{\rho-1}Y^{1-\rho} \quad (2)$$

$$p_I = (1-a)Y_I^{\rho-1}Y^{1-\rho}. \quad (3)$$

It is important to note that both marginal productivities depend on the relative importance of the formal product in the production of the final good (parameter a). Thus, although firms are homogeneous ex ante, formal and informal firms might have different productivities ex post. In fact, I make the assumption that once firms decide to produce in the informal sector they suffer a productivity loss, which is equivalent to assume that $0.5 < a < 1$. This hypothesis can be justified by at least two factors. First, informal firms cannot fully exercise property rights over their products and capital, which implies higher transaction costs and difficulties in accessing credit. Second, government control may prevent informal firms from growing and, therefore, from exploiting possible economies of scale.

The only factor used to produce the intermediate good is labor. Each firm hires one, and only one, worker who produces exactly one unit of the good. There is a continuum of homogeneous and risk-

¹² Schneider and Enste (2000) and Schneider and Klinglmaier (2004) present estimates of the size of the informal sector around the world.

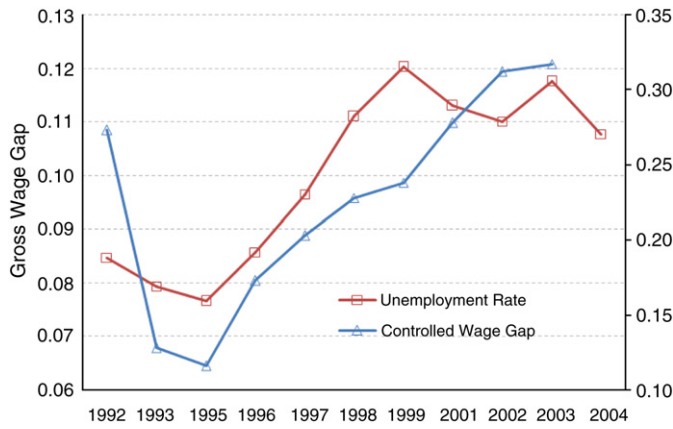


Fig. 3. Controlled formal-informal wage gap and unemployment.

neutral workers with measure one and a larger continuum of firms that are identical and risk-neutral.¹³ The discount rate is the same for workers and firms and is denoted by r . When firms decide to produce they must choose in which sector they are going to enter and, therefore, what type of vacancy they are going to create.

To create a formal job, firms must incur a creation cost K_F while the cost to create an informal one is given by K_I , where $K_F > K_I$. The informal sector is often characterized as being less capital intensive than the formal sector and hence this assumption could be justified exclusively through technological differences. However, I take a different route here and assume that the difference between creation costs in both sectors is entirely due to institutional factors: in order to create a formal vacancy, the firm has to incur in bureaucratic and monetary costs that come from the existence of a (potentially restrictive) regulation of entry.¹⁴ Thus, I assume that the capital requirement per vacancy created is the same in both sectors.

Even though this assumption might seem strong at first glance, it is worth noting that informal employment in developing countries is highly concentrated in the services sector, where indeed capital requirements do not vary greatly across firms. Furthermore, this assumption does not alter any of the results and all the discussion in the paper holds if instead I assume that part of the difference between K_F and K_I is due to structural differences between both sectors. The reason this assumption is made is simply to emphasize the importance of institutional factors in determining the cost of entry into the formal sector and its consequences to labor market performance, even in a context of no differences in capital requirements in both sectors.

The existence of technological and institutional factors that distinguish both sectors implies that even when the latter are removed (and therefore there are no policy distortions), the informal sector does not disappear entirely. Formal and informal production are not perfect substitutes ($\rho < 1$) and the production function of the final good places a positive weight on informal production ($a < 1$). This means that there is some need for diversification in the production of the final good (see Acemoglu, 2001) and the informal sector will not disappear in the absence of institutions and regulations.

However, in this extreme case the distinction between formal and informal is no longer informative, as there are no institutions to comply with. Put it differently, the informal sector is usually defined as

all activities that are legal but that do not comply with government's regulations; hence, once we remove all labor and entry regulation the informal sector ceases to exist by definition. In this extreme scenario, instead of having one sector that complies and one that does not comply with government's regulation, my model would have simply two sectors producing intermediate goods that are imperfect substitutes in the production of the final good.¹⁵ Thus, this model is not adequate to analyze how the informal sector is created, as it is introduced in a somewhat arbitrary way (as it is the case in almost all the literature). Instead, this model is useful to analyze the impact of policy changes given the existence of an informal sector.

3.2. The labor market

The trade process in the labor market is summarized by two sectorial matching functions denoted by $m_j(u, v_j)$, where u is the aggregate unemployment rate and v_j is the vacancy rate in sector $j = I, F$.¹⁶ The function $m_j(u, v_j)$ is assumed to be twice differentiable, increasing in its arguments and to exhibit constant returns to scale. There is no on-the-job search. Because of constant returns to scale, it is possible to write the flow rate for firms as $q_j(\theta_j) = \frac{m_j(u, v_j)}{v_j}$, where $q_j(\cdot)$ is a differentiable and decreasing function, and θ_j denotes labor market's tightness in sector j . The tightness is determined by the ratio between the vacancy rate in sector j and the aggregate unemployment rate. Thus, this parameter represents labor demand and supply conditions prevailing in sector j . Similarly, the flow rate for workers is given by $\lambda_j(\theta_j) = \frac{m_j(u, v_j)}{u} = \theta_j q_j(\theta_j)$. Assuming that $q_j(\theta_j) > 0$, $\theta_j q_j(\theta_j) < \infty$ means that finding a job or an employee is a time-consuming activity and that it takes time for workers and firms to make an adequate match. The usual Inada-type conditions on $m_j(u, v_j)$ apply, so that $\lim_{\theta_j \rightarrow 0} q_j(\theta_j) = 0$, $\lim_{\theta_j \rightarrow \infty} q_j(\theta_j) = \infty$, $\lim_{\theta_j \rightarrow 0} \lambda_j(\theta_j) = 0$ and $\lim_{\theta_j \rightarrow \infty} \lambda_j(\theta_j) = \infty$.¹⁷

Formal and informal jobs end at a constant exogenous separation rate denoted by s_F and s_I , respectively, with $s_I > s_F$. The assumption that labor relations are more unstable in the informal sector (higher separation rate) is supported by the evidence presented in Table 2 (Section 2) and in several studies documenting that informal jobs have a shorter spell and present higher turnover rates (among others, Maloney (1999)). When a separation occurs the firm becomes an unfilled vacancy and the worker becomes unemployed.

Displaced workers coming from both sectors constitute a pool of undifferentiated unemployed who search for jobs in both sectors with the same intensity. Hence, the matching function's argument is the aggregate unemployment rate. This constitutes an innovation relatively to the previous literature, as it combines separated markets and undirected search. Separated markets allow me to analyze formal and informal sectors separately and therefore I can compute separated tightness, vacancy rates and employment probabilities. Undirected search implies that workers can apply to both sectors simultaneously and there is no need to rely on sectorial unemployment rates. Typically in the literature these two features appear separately in the models, so one has to choose between the two approaches; the approach adopted here relaxes this tradeoff. In addition to that, this

¹³ The labor force is supposed to be constant what is equivalent to assume that workers are infinitely lived.

¹⁴ Djankov et al. (2002) document and analyze the regulation of entry and the costs associated to it in 85 countries.

¹⁵ If I relax the assumption that all the difference in the cost of creating a vacancy in both sectors is driven by institutional factors and allow for differences in capital requirements to play a role, then the analysis under this extreme scenario becomes very similar to the one in Acemoglu (2001). The only difference would be the innovation in the matching structure that is introduced in Section 3.2.

¹⁶ The vacancy rate is determined by the ratio between the number of vacancies created in each sector and total labor force.

¹⁷ To make the quantitative experiments, it is necessary to assume a functional form for $m_j(u, v_j)$. I use the standard Cobb–Douglas matching function that is given by $m_j(u, v_j) = A(v_j)^{1-\eta} (u)^\eta$, $j = I, F$. Therefore, the probability of an unemployed worker to get a job in sector j is given by $\lambda_j(\theta_j) = A(\theta_j)^{1-\eta}$ and for a firm holding an open vacancy on sector j , the probability of finding a worker is given by $q_j(\theta_j) = A(\theta_j)^{-\eta}$, $j = I, F$.

structure creates a straight link between formal and informal sectors in the sense that labor supply and demand conditions in one sector directly affect agent's decisions in the other sector.

3.3. Firms and workers characterization

Workers and firms are completely characterized by equations that depict the present discounted value of each possible status in the labor market. Firms face two possible situations to be a filled or an unfilled vacancy while workers can be unemployed, employed in the formal sector or employed in the informal sector. The equations of the present discounted value of a filled vacancy in formal and informal sectors, respectively, and of an unfilled vacancy in both sectors are the following:

$$rJ_F = p_F - (1 + \tau_\pi)w_F - s_F(J_F - V_F) \quad (4)$$

$$rJ_I = p_I - w_I - s_I(J_I - V_I) \quad (5)$$

$$rV_j = q_j(J_j - V_j), \quad j = I, F. \quad (6)$$

The intuition behind these equations, as for the other value equations in this model, is that holding a filled or unfilled vacancy in any sector corresponds to holding an asset. In the case of a filled vacancy in the formal sector (rJ_F), it pays a “dividend” of p_F but there is a cost of keeping this asset that is equal to total formal labor cost the basic contractual wage (w_F) plus the payroll tax (τ_π). Additionally, there is a cost related to the uncertainty of the asset's return reflected by the probability s_F that the match will be terminated, in which case the firm will suffer a net loss of $J_F - V_F$. Finally, I suppose free-entry on the firm side, so it is not possible to open an additional vacancy and make positive expected net profits. Thus, we have that $rV_j = rK_j$ (free-entry assumption).

On the workers' side, the value equations depict the present discounted value of being employed in the formal and informal sectors, respectively, and of being unemployed (common to all workers). Respectively, the equations are the following:

$$rE_F = (1 - \tau_\omega)w_F - s_F(E_F - U) + s_F b \quad (7)$$

$$rE_I = w_I - s_I(E_I - U) \quad (8)$$

$$rU = \lambda_F(E_F - U) + \lambda_I(E_I - U) \quad (9)$$

where τ_ω is the labor tax and b is the present discounted value of the total expected flow of unemployment benefits. Therefore, the value of being employed in both sectors is equal to net total earnings, which in the case of a formal worker includes the total expected unemployment benefit (b) weighted by the probability of becoming unemployed (s_F). Workers in both sectors face a positive probability of becoming unemployed that implies a net loss of $(E_j - U)$, $j = I, F$. Finally, as there is a pool of undifferentiated unemployed workers and no directed search, the value equation of being unemployed is the same for all workers. It is given by the sum of net gains of entering in both sectors weighted by their respective entry probability (λ_j).

There is a rent sharing process between workers and firms over the total surplus of the match, which is modeled using a standard Nash bargaining framework. As firms and workers are risk-neutral and have the same discount factor, the first order conditions of the bargain imply that formal and informal wages must respect the following first order conditions:

$$(1 - \phi_j)(E_j - U) = \phi_j(J_j - V_j), \quad j = I, F \quad (10)$$

where ϕ_j denotes workers' bargaining power in sector j .

3.4. Wage equations and arbitrage conditions

Using the value equation of employment in both sectors (Eqs. (7) and (8)), the first order conditions of the Nash bargaining (Eq. (10)) and the free-entry condition ($rV_j = rK_j$), it is possible to obtain the wage equation for the formal and informal sectors:

$$w_F = \frac{1}{1 - \tau_\omega} \left[\frac{\phi_F}{1 - \phi_F} \frac{(r + s_F)}{q_F} rK_F + rU - s_F b \right] \quad (11)$$

$$w_I = \frac{\phi_I}{1 - \phi_I} \frac{(r + s_I)}{q_I} rK_I + rU. \quad (12)$$

In order to completely determine these wage equations, it is necessary to define the value equation of unemployment (rU). Using Eq. (9), Nash bargaining's first order conditions and free-entry assumption:

$$rU = \frac{\phi_F}{1 - \phi_F} rK_F \theta_F + \frac{\phi_I}{1 - \phi_I} rK_I \theta_I. \quad (13)$$

On the firm side, the arbitrage conditions are obtained using the value equation of a filled vacancy in both sectors (Eqs. (4) and (5)), as well as the free-entry condition:

$$p_F = (1 - \tau_\pi)w_F + rK_F + (r + s_F) \frac{rK_F}{q_F} \quad (14)$$

$$p_I = w_I + rK_I + (r + s_I) \frac{rK_I}{q_I}. \quad (15)$$

3.5. Steady state equilibrium

Both sectors exist in equilibrium. As each worker produces only one unit of the intermediate good, the aggregate production of formal and informal sectors is equal to the number of workers employed in each sector. Defining γ as the proportion of formal jobs in the economy, the aggregate production in formal and informal sectors is given by: $Y_F = \gamma(1 - u)$ and $Y_I = (1 - \gamma)(1 - u)$. Thus, from Eqs. (2) and (3):

$$p_F = a\gamma^{\rho-1}(a\gamma^\rho + (1-a)(1-\gamma)^\rho)^{\frac{1-\rho}{\rho}} \quad (16)$$

$$p_I = (1-a)(1-\gamma)^{\rho-1}(a\gamma^\rho + (1-a)(1-\gamma)^\rho)^{\frac{1-\rho}{\rho}}. \quad (17)$$

Steady state equilibrium is characterized by the two equilibrium loci that result from the substitution of the wage equations (Eqs. (11) and (12)) in the arbitrage conditions (Eqs. (14) and (15)):

$$p_F = \frac{(1 + \tau_\pi)}{1 - \tau_\omega} \left[\frac{\phi_F(r + s_F)}{(1 - \phi_F)q_F} rK_F + rU - s_F b \right] + (r + s_F + q_F) \frac{rK_F}{q_F} \quad (18)$$

$$p_I = \frac{(r + s_I)}{(1 - \phi_I)q_I} rK_I + rU + rK_I. \quad (19)$$

Finally, no sector can be expanding or contracting in equilibrium, so the number of jobs destroyed must equal flows out of unemployment (Pissarides, 2000). This gives two equilibrium conditions, which determine the flow condition that must hold in equilibrium:

$$\gamma = f(\theta_F, \theta_I) = \frac{\lambda_F s_I}{\lambda_F s_I + \lambda_I s_F}. \quad (20)$$

With this last condition it is possible to completely characterize the model's equilibrium. Equating equilibrium marginal productivities (Eqs. (16) and (17)) to their respective equilibrium locus (Eqs. (18) and (19)) results in two equations that implicitly define γ as a function of θ_F and θ_I . These two equations, together with the flow condition (Eq. (20)), form a system of three equations and three unknowns (γ , θ_F and θ_I). This system can be further simplified: substituting the flow condition in Eqs. (16) and (17), it is possible to obtain expressions for p_F and p_I (denoted by $\tilde{p}_F$ and $\tilde{p}_I$) as functions exclusively of both sector's tightness: $\tilde{p}_F = g_F f(\theta_F, \theta_I)$ and $\tilde{p}_I = g_I (f(\theta_F, \theta_I))$. Using these two expressions and equilibrium loci of formal and informal sectors, it is possible to write the following system:

$$\begin{aligned} p_F - \tilde{p}_F &= 0 \\ p_I - \tilde{p}_I &= 0. \end{aligned} \quad (21)$$

This system is completely nonlinear over the variables θ_F and θ_I , which makes it very difficult to obtain an analytical solution or to prove the existence of equilibrium. Nonetheless, it is possible to show that the equilibrium exists and it is unique for a broad range of parameters. Section 3.5.3 discusses the properties of the model's steady state equilibria.

3.5.1. Wage differentials in steady state equilibria

One of the aspects largely discussed in the informal sector literature is the one regarding formal–informal wage differentials.¹⁸ In the present model, the equilibrium wage gap is given by the following expression:

$$w_F - w_I = \frac{1}{1 - \tau_\omega} \left[\frac{\phi_F(r + s_F)rK_F}{(1 - \phi_F)q_F} - s_F b \right] - \frac{\phi_I(r + s_I)rK_I}{(1 - \phi_I)q_I} + \frac{rU\tau_\omega}{1 - \tau_\omega}. \quad (22)$$

As can be seen, wage differences are completely determined by institutional parameters and labor market conditions in both sectors (given by the market tightness). Contrary to most of the literature where the wage gap is always positive in favor of formal sector workers, here the wage gap has an ambiguous sign. This gives the model considerable flexibility, as the direction of the wage gap is not determined *a priori*.

Looking at partial equilibrium effects (holding fixed labor market tightness), it is straightforward to see that an increase in the cost of creating a formal vacancy (K_F) causes the wage gap to increase. A higher entry cost implies that a formal match needs to generate a higher surplus in order to compensate these higher costs; as there is rent sharing between workers and firms, formal wage increases. This can be seen more clearly if we make the following simplifying assumptions: $\tau_\omega = b = 0$; $\phi_F = \phi_I = 1/2$; and $s_F = s_I = s$. Under these assumptions, the wage gap is given by

$$w_F - w_I = (r + s) \left(\frac{rK_F}{q_F} - \frac{rK_I}{q_I} \right). \quad (23)$$

This expression shows that when the only institutional difference between both sectors is the cost of creating a vacancy ($K_F > K_I$), the wage gap is positive as long as the probability of finding a worker in the formal sector (q_F) is not much higher than that in the informal sector (or the tightness in the formal sector is not much lower than that in the informal sector). Thus, even in this simpler case the sign of the wage gap depends on the tightness in both sectors (θ_F and θ_I).¹⁹

¹⁸ As shown in Section 2, the controlled wage gap in Brazil has been quite high over the past ten years: observably equivalent works are paid on average 30% more just for being formal. These naive estimates are evidently biased for they do not account for self-selection, but even when one tries to correct for endogeneity the differential remains high.

¹⁹ It is worth noting that this is true provided that there are market frictions that create rents to be shared by workers and firms. Once these frictions disappear – that is, when $q_j \rightarrow \infty$, $j = F, I$ – the wage gap vanishes.

Unemployment benefits work as subsidy to formal employment, as the benefits are perceived by formal workers as a lump sum income (discounted by the probability of losing the formal job, s_F), so an increase in b leads to a decrease in formal wages. As expected, an increase in payroll taxes leads to an increase in formal wages and thus to an increase in the wage gap (all else constant). Finally, an increase in the informal sector separation rate leads to a decrease in the wage gap; the intuition is that with a higher probability of losing their jobs once they join the informal sector, workers will require a higher wage in order to accept an informal job, thus decreasing the wage gap.

3.5.2. Welfare

Before proceeding to the quantitative experiments, it is important to define a measure of the economy's welfare in order to assess the net impact of the policy experiments that are performed. Following Acemoglu (2001), I measure welfare as the steady state net surplus of the economy, given by the total output net of the costs of creating all the vacancies in the economy:

$$W = (1 - u)[\gamma(p_F - rK_F) + (1 - \gamma)(p_I - rK_I)] - u(\gamma\theta_F rK_F + (1 - \gamma)\theta_I rK_I). \quad (24)$$

One important question is whether changes in the composition of jobs, holding constant the tightness in both sectors, have a positive or negative impact on the level of welfare. Differentiating the above expression with respect to the share of formal jobs:²⁰

$$\frac{\partial W}{\partial \gamma} = (1 - u) \left[(p_F - p_I) + \gamma \frac{\partial p_F}{\partial \gamma} + (1 - \gamma) \frac{\partial p_I}{\partial \gamma} - (rK_F - rK_I) \right] - u(\theta_F rK_F - \theta_I rK_I). \quad (25)$$

Using equilibrium Eqs. (16) and (17), it is possible to show that for the values of ρ and a used in the parameter configuration for the baseline economy (Section 4), we have that $\frac{\partial p_F}{\partial \gamma} < 0$ and $\frac{\partial p_I}{\partial \gamma} > 0$.

To gain more intuition about the mechanisms that are behind the numerical results, it is useful to consider again the case where there are no institutional differences between both sectors except that $K_F > K_I$.²¹ In this case and using Eqs. (18) and (19) we have that

$$p_F - p_I = \frac{r + s}{1 - \phi} \left(\frac{rK_F}{q_F} - \frac{rK_I}{q_I} \right) + (rK_F - rK_I). \quad (26)$$

Substituting this equation into expression (25) we get the following:

$$\frac{\partial W}{\partial \gamma} = (1 - u) \left[\frac{r + s}{1 - \phi} \left(\frac{rK_F}{q_F} - \frac{rK_I}{q_I} \right) + \gamma \frac{\partial p_F}{\partial \gamma} + (1 - \gamma) \frac{\partial p_I}{\partial \gamma} \right] - u(\theta_F rK_F - \theta_I rK_I). \quad (27)$$

Thus, we can see that the difference in creation costs has two opposite effects on the above expression (for given tightness levels). On the one hand, it enhances the price difference between the two sectors, which makes it more likely that $\frac{\partial W}{\partial \gamma} > 0$. The reason for that is that if p_F is higher than p_I , increasing the share of formal jobs has a higher gross benefit than increasing the share of informal jobs. On the other hand, there is a negative impact that comes from the unfilled vacancies: the greater the difference between K_F and K_I , the higher the negative term associated to the deadweight loss of unemployment (last term in the RHS of Eq. (27)).

Clearly, which effect dominates depends on the tightness in both sectors and on the level of unemployment. In particular, higher levels

²⁰ To compute this derivative I hold constant the tightness in both sectors. To simplify the analysis, I evaluate this derivative around the steady state, so that I can use equilibrium relationships for the prices (Eqs. (16)–(19)). I also use the fact that the equilibrium unemployment does not depend on the share of formal jobs, but only on the separation rates and tightness in both sectors.

²¹ As in the previous section, I assume that $\tau_\omega = b = 0$; $\phi_F = \phi_I = \phi$; and $s_F = s_I = s$.

of unemployment combined with a large difference in K can cause increases in the share of formal jobs to have a negative effect on welfare. This is the case observed in the numerical exercise of increases in the level of enforcement discussed in Section 4. In this simulation, I increase the informal separation rate but hold all the remaining institutional parameters at their baseline levels, and so the difference between K_F and K_I is kept at a high level. Raising the informal separation rate leads to significant increases in both the unemployment level and in the share of formal jobs, which is accompanied by a substantial decrease in welfare, as the above analysis would suggest.

3.5.3. Uniqueness of equilibria

As commented before, drawing a richer portrait of the labor market structure and its institutions comes at the cost of greater parameter dimensionality and less analytical tractability. One major concern in this context is about the uniqueness of steady state equilibria. Thus, a careful numerical analysis of the properties of the steady state equilibria is needed.

Formal and informal equilibrium loci (Eq. (21)) implicitly define two expressions that give θ_I as a function of θ_F , say, $\theta_I^F \equiv \xi_F(\theta_F)$ and $\theta_I^I \equiv \xi_I(\theta_F)$. Steady state equilibrium is then determined by the intersection of these two curves in the θ_I – θ_F plane. Thus, to analyze the properties of steady state equilibria it suffices to study the plot of these two equilibrium loci for different combinations of parameters. Here, I consider the baseline economy and all parameter configurations that correspond to the policy experiments discussed in Section 4. Additionally, I analyze the case of no institutional differences between both sectors, where: there are no taxes, the cost of creating a vacancy is the same in both sectors (and equal to the cost in the informal sector), no unemployment benefits, same separation rates ($s_F = s_I$) and same workers' bargaining power in both sectors. Finally, I also consider different values of the elasticity parameter of the CES production function: $\rho = \{0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$.

As the graphs in the Appendix show, for the baseline economy and all set of parameters considered in the policy experiments the model presents an unique equilibrium (the curves θ_I^F and θ_I^I intersect only once in the θ_I – θ_F plane). Turning to the different values of ρ , the graphs show that for $\rho \in \{0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7\}$ (maintaining the remainder of the parameters at their baseline specification) the model still yields a unique equilibrium.

However, as $\rho \rightarrow 1$ the model starts presenting multiple solutions (more precisely, there are two possible equilibria for $\rho = .8$ and $\rho = .9$). When $\rho = 1$ and we have the case where formal and informal production are perfect substitutes, the model presents a continuum of corner solutions along the θ_I axis. This is consistent with the analytical example given by Acemoglu (2001). Nonetheless, as argued in Section 3.1, this extreme scenario where formal and informal production are perfect substitutes is highly implausible, and does not seem to be consistent with the data. The same could be argued about the scenarios where $\rho = .8, .9$. Even though they do not correspond to the case of perfect substitutes, they imply a very high degree of substitutability. Moreover, it is much higher than the usual estimated value of ρ found in the literature, which is around 0.3 (among others, Fernandes et al. (2005)).

Finally, a possible albeit unlikely parameter configuration that can lead to multiple equilibria is the one with no institutional differences between both sectors. This corresponds to the case where we eliminate all taxes and unemployment benefits, and equalize workers' bargaining power, separation rates and entry costs across sectors. This is also an extremely unfeasible scenario, which can nonetheless generate more than one steady state equilibrium.

Thus, although it is not possible to prove uniqueness formally, this numerical analysis provides satisfactory evidence that the model presents a unique equilibrium over a broad range of parameters, which increases the reliability of the results presented in Section 4.

4. Quantitative experiments

4.1. Calibration

The quantitative exercises consist in solving the equation system presented at the end of the previous section (Eq. (21)) for each set of new parameters that characterize the policy experiments. The simulation exercises are restricted to parameters related to institutions, while the remaining are treated as structural parameters of the economy and, therefore, remain unchanged. Some of the parameters are directly observed in the data, others can be estimated or extracted from previous empirical works, and a small share of them must be calibrated within the model. Table 3 presents the set of initial parameters that characterize the status quo of the Brazilian labor market.

The value adopted for formal workers' bargaining power is the standard value in the literature. For informal workers, I use the data presented by Camargo (2002) to get an approximation for informal workers' bargaining power, which is set to be one third that of formal worker's. In addition to the numerical analysis presented in Section 3.5.3, many robustness tests were done in order to access the sensibility of the results with respect to parameters ϕ_F and ϕ_I . The results revealed to be quite robust: only the level of the variables is affected, while all relative results hold independently of what value is adopted. The parameters A , η and a are calibrated so the model's equilibrium gives the unemployment rate, formal and informal employment rates observed in the 2005 National Household Survey (Pesquisa Nacional por Amostra de Domicílios).

It is important to note that these labor market indicators are computed taking into account the population definition used in the present model, so the total labor force is equal to the sum of unemployed, formal and informal workers, which encompasses informal salaried workers and the self employed. The informal sector is computed as the share of informal workers over this restricted population, and the same applies to formal employment and unemployment rates. In addition to these, some other indicators are used to guide the calibration procedure, mainly the controlled wage differential between formal and informal workers.

4.2. Results

Given this baseline specification, the following experiments are performed: (i) variations in unemployment benefits; (ii) variations in formal sector entry costs; (iii) increases in enforcement of labor market institutions (through increases in s_I , the separation rate); and (iv) variations in payroll tax. Due to the large number of results, each

Table 3
The set of initial parameters – the baseline economy.

Parameters	Value	Source/rationale
Payroll tax (τ_n)	35%	Inter-Union Depart. of Statistics
Labor tax (τ_w)	10%	Social security contribution
Discount rate (r)	8%	Heckman and Pagés (2000)
Formal bargain power (ϕ_F)	45%	Standard value in the literature
Informal bargain power (ϕ_I)	15%	Estimates from Camargo (2002)
Informal separation rate (s_I)	30%	Literature and Table 2
Formal separation rate (s_F)	13%	Literature and Table 2
CES elasticity parameter (ρ)	0.3	Estimates from the literature
Formal sector's entry cost (K_F)	1.4	Estimated ^a
Informal sector's entry cost (K_I)	0.30	Estimated ^a
Unemployment benefit (b)	0.30	Estimated ^a
Matching elasticity (η)	0.5	–
Formal sector's relevance (a)	0.62	–
Matching scale parameter (A)	0.30	–
Unemployment rate (u)	13.7%	National Household Survey 2005
Informal sector [$(1 - \gamma)(1 - u)$]	45.37%	National Household Survey 2005
Formal sector [$\gamma(1 - u)$]	41.36%	National Household Survey 2005

^a Parameter estimated following the methodology from Heckman and Pagés (2000).

Table 4
Variation in formal sector's entry costs.

	$K_F = 0.4$	$K_F = 0.8$	$K_F = 1.4^a$	$K_F = 2.0$	$K_F = 2.5$
Unemployment (u)	8.5	10.9	13.3	15.0	16.1
Formal sector [$\gamma(1-u)$]	54.2	47.8	41.4	36.6	33.4
Informal sector [$(1-\gamma)(1-u)$]	37.3	41.3	45.4	48.4	50.5
Wage differential (in %)	13.0	22.1	32.3	40.6	46.7
Average productivity	1.044	1.027	1.000	0.971	0.947
Welfare	1.257	1.138	1.000	0.890	0.813
Pr(unemp. → formal sector) (λ_F)	82.5	56.9	40.5	31.8	27.0
Pr(unemp. → informal sector) (λ_I)	90.9	73.6	62.6	57.1	54.2
Formal vacancy filling rate (q_F)	10.9	15.8	22.2	28.3	33.3
Informal vacancy filling rate (q_I)	6.9	7.9	8.8	9.3	9.6

^a Initial value.

experiment is analyzed separately and each subsection presents a summary table containing the model's main results.

4.2.1. Entry costs

I start by analyzing the impacts of reducing formal sector's entry costs, which are presented on Table 4. As one can immediately observe, the cost of creating a formal vacancy has a substantial impact on all the model's outcomes.

The effects on employment composition are particularly strong. Consider the case of bringing formal entry costs close to the level of informal sector's, which is equivalent to eliminating all bureaucratic and monetary costs associated to opening a formal vacancy: reducing K_F from its baseline value to 0.4 would generate an increase of nearly 13 percentage points (p.p.) in formal employment share and would reduce informality by 8.1 p.p. Unemployment would also be affected, declining 4.8 p.p. relative to its baseline value. This reduction on unemployment is due mainly to the sharp increase in exit probabilities from unemployment to both sectors, with the increase being relatively much higher in the exit rate to the formal sector. At the same time, formal–informal wage gap would be reduced to nearly one-third of its baseline value, which is a result of two concomitant effects: (i) formal sector employment is increasing and therefore formal workers' marginal productivity must be falling; (ii) higher entry costs imply that formal matches need to generate higher surplus, and the rent sharing between workers and firms leads to higher formal wages.²² When entry costs are reduced, this latter mechanism becomes less important and the wage gap decreases.

As a consequence of the changes on employment composition and the reduction of unemployment rate, average productivity and specially welfare show a substantial increase. Hence, high entry costs not only play a major role in determining employment composition, but also generate a considerable deadweight loss. This latter effect comes from the fact that high creation costs reduce the number of formal vacancies created and, as the informal sector is not able to completely absorb all workers, unemployment increases.

In light of these results, the political economy argument that the informal sector is largely tolerated because eradicating it would significantly increase unemployment becomes less appealing. Put differently, the results presented here show that the tradeoff between lower informal employment and higher unemployment rates does not exist when one considers policies that reduce the cost of being formal, in opposition to policies that simply increase the cost of being informal.

These results also show that unemployment increases that come from increases in the start up costs in the formal sector are accompanied by increases in the wage gap. Thus, the model's results are in line with the empirical relationship between the controlled wage gap

²² For a more careful discussion, see Acemoglu (2001). Similarly, Auriol and Warlters (2005) show that higher entry costs generate market power for the formal firms that imply higher rents to be shared among the agents.

Table 5
Changes in payroll tax.

	$\tau_\pi = 0.15$	$\tau_\pi = 0.25$	$\tau_\pi = 0.35^a$	$\tau_\pi = 0.50$	$\tau_\pi = 0.70$
Unemployment (u)	13.1	13.2	13.3	13.4	13.6
Formal sector [$\gamma(1-u)$]	43.5	42.4	41.4	40.0	38.3
Informal sector [$(1-\gamma)(1-u)$]	43.4	44.4	45.4	46.6	48.1
Wage differential (in %)	34.1	33.2	32.3	30.9	29.1
Average productivity*	1.012	1.006	1.000	0.991	0.979
Welfare	1.013	1.005	1.000	0.991	0.978
Pr(unemp. → formal sector) (λ_F)	43.3	41.9	40.5	38.7	36.6
Pr(unemp. → informal sector) (λ_I)	59.7	61.2	62.6	64.3	66.0
Formal vacancy filling rate (q_F)	20.8	21.5	22.2	23.2	24.6
Informal vacancy filling rate (q_I)	9.0	8.9	8.8	8.6	8.5

^a Initial value.

and unemployment rate found since mid 1990s (Section 2). Moreover, during this period fixed costs of hiring formally increased substantially in Brazil as a consequence of the constitutional change in 1988 and subsequent changes in labor legislation (Gonzaga, 2003). Hence, these simulation results offer a rationale for the aggregate behavior observed in the labor market during the 1990s and early 2000s.

4.2.2. Payroll tax and unemployment benefits

Considering first the impact of varying the payroll tax, Table 5 clearly shows that either an increase or a decrease in τ_π has limited effects on the size of the informal sector and on unemployment rate: cutting the tax by 57.1% (from 35% to 15%) reduces the size of the informal sector by only 2 p.p., while unemployment rate remains practically stable. The same can be said about exit probabilities out of unemployment to both sectors, as we can only observe a minor change in both λ_F and λ_I . The effects on the wage gap are not substantial either, which supports the view that payroll tax reductions tend to have a very reduced effect on wages in the long run.²³ As a result of these mild effects on labor market outcomes, average productivity and welfare show only a slight improvement.

These results are in line with the empirical findings of Botero et al. (2004). The authors find no statistically significant effect of both employment protection index and social security index on informal employment. Albrecht et al. (2009) find similar compositional effects but with a somewhat greater magnitude as the ones presented in Table 5. However, they find that the net output plus total tax revenues increases with the payroll tax, which is the opposite of the result presented here. The reason for this divergence is the fact that the authors assume that increasing the size of the unemployment pool does not affect the probability of finding an informal job; this creates a congestion externality in the formal labor market that can be reduced by the introduction of a payroll tax. The present model does not generate this result because of the modification in its matching structure, which makes both sectors more integrated. Hence, the model is able to capture the distortion generated by increasing payroll taxes.

Turning to the unemployment benefit, one first interesting result presented in Table 6 is its positive, albeit very reduced, effect on unemployment rate. Therefore, the benefit does not show its standard adverse effect of increasing unemployment; on the contrary, it leads to a small increase in exit probability out of unemployment to formal employment. However, since it also leads to a decrease in exit probabilities to informal employment, the overall effect on unemployment is almost null.

²³ Nonetheless, the results in the empirical literature regarding the effects of payroll taxes on wages and employment are mixed (see Kugler and Kugler, 2009).

Table 6
Changes in the unemployment benefit.

	$b = 0$	$b = 0.2$	$b = 0.3^a$	$b = 0.6$
Unemployment (u)	13.5	13.3	13.3	13.1
Formal sector [$\gamma(1-u)$]	39.4	40.7	41.4	43.3
Informal sector [$(1-\gamma)(1-u)$]	47.2	46.0	45.4	43.6
Wage differential (in %)	51.4	38.4	32.3	15.9
Average productivity*	0.985	0.995	1.000	1.013
Welfare	0.987	0.996	1.000	1.011
Pr(unemp. $\rightarrow$ formal sector) (λ_F)	38.0	39.7	40.5	43.0
Pr(unemp. $\rightarrow$ informal sector) (λ_I)	64.9	63.4	62.6	59.9
Formal vacancy filling rate (q_F)	23.7	22.7	22.2	20.9
Informal vacancy filling rate (q_I)	8.6	8.7	8.8	9.0

^a Initial value.

The impacts in employment composition are also very limited, with a slight reduction of informal sector's size and an expansion of formal employment. Again, these effects come from the fact that the unemployment benefit works as a subsidy to hiring formal workers. From Eq. (7), one can observe that the benefit enters as a lump sum income, which is deducted from formal workers' equilibrium wage (Eq. (11)). This subsidy characteristic becomes clearer by observing the extremely strong effects on the wage differential between formal and informal workers: increasing the benefit from 0.3 to 0.6 would halve the wage gap.²⁴

4.2.3. Enforcement

Increasing enforcement of current labor market institutions (by means of imposing higher separation rates) has large effects in labor market outcomes, specially employment composition and unemployment rate, as shown in Table 7. Increasing informal sector's separation rate from 0.3 to 0.7 would cause a 14.3 p.p. decrease in the size of the informal sector, which would more than compensate the 13 p.p. growth observed during the past decade in Brazil (Fig. 1). However, formal employment would only increase 8.5 p.p., resulting in a substantial increase in unemployment (nearly 6 p.p.). The probability of exiting unemployment to a formal job would decrease by 15%, while the exit rate to informal employment would increase. An interesting effect is the significant drop in the wage gap; it reflects the fact that when facing higher separation rates, informal firms must offer higher wages in order to attract workers to informal jobs, which are now much riskier than formal ones.

It is also worth noting that in this scenario we can observe an inverse relationship between wage differentials and unemployment rate, and a negative correlation between unemployment and informal sector's size. However, as discussed in Section 2, this scenario is not compatible with Brazil's labor market behavior in the 1990s and mid 2000s: during this period the informal sector increased and so did the unemployment rate; additionally, the controlled wage gap and the unemployment rate displayed a positive correlation. Thus, these simulation results suggest that changes in the enforcement level of labor market institutions was not the cause of the observed changes in Brazilian labor market outcomes in the 1990s and mid 2000s.

Finally, all the effects combined result in a total productivity increase of 4.7% and in a significant decrease in welfare level (14.7%). The productivity increase simply reflects the changes in employment composition: as the formal sector is more productive ($\alpha > 0.5$), the increase in the share of formal jobs leads to an increase in average productivity. However, this improvement comes together with a

Table 7
Increasing institutions' enforcement.

	$s_I = 0.3^a$	$s_I = 0.5$	$s_I = 0.7$
Unemployment (u)	13.3	16.5	19.1
Formal sector [$\gamma(1-u)$]	41.4	46.4	49.9
Informal sector [$(1-\gamma)(1-u)$]	45.4	37.2	31.1
Wage differential (in %)	32.3	22.0	13.3
Average productivity*	1.000	1.034	1.047
Welfare	1.000	0.921	0.853
Pr(unemp. $\rightarrow$ formal sector) (λ_F)	40.5	36.5	34.0
Pr(unemp. $\rightarrow$ informal sector) (λ_I)	62.6	72.6	74.1
Formal vacancy filling rate (q_F)	22.2	24.6	26.4
Informal vacancy filling rate (q_I)	8.8	8.0	7.9

^a Initial value.

significant unemployment increase, which is reflected by the substantial drop in the welfare level. Thus, these results show that reducing the size of the informal sector cannot be an objective in itself, as a reduction in the size of the informal sector can be accompanied by a general worsening of labor market performance.

In the present context, this adverse overall effect is a result of a more intensive enforcement of existing restrictive labor market institutions and high entry costs in the formal sector. These results offer a rationalization for the so-called “shadow puzzle”. Increasing the compliance with the current inefficient and excessive labor and entry regulation apparatus by simply intensifying the degree of coercion of informal activities would lead to higher unemployment; therefore the informal sector is tolerated even though it would be possible to significantly reduce it by imposing more severe restrictions onto its existence.

5. Final remarks

This paper examines the role of labor market institutions and the regulation of entry on the size of the informal sector and on overall labor market performance. For that, I develop a two-sector matching model that includes the main characteristics of developing countries' labor and entry regulations. The model presents an integrated approach, as it considers the interactions between institutions and the decisions of workers and firms to supply and demand labor in both sectors. It presents an important innovation in its matching structure, as it combines separated markets and undirected search. This innovation makes the model more coherent with some key stylized facts of the informal sector in developing countries. The model is parameterized using the Brazilian economy, which is a highly regulated developing country (considering both labor and entry regulations) and that has an extremely large informal sector (nearly 50% of total employment). Additionally, the informal sector in Brazil has been steadily increasing for almost 15 years and the observed upward trend started just after a major institutional change in 1988 that significantly increased labor costs.

The simulation results indicate that a crucial determinant of the size of the informal sector is the magnitude of entry costs into the formal sector. High entry costs are associated with higher informality, lower welfare and worse overall labor market performance: higher unemployment rates (lower exit rates from unemployment), higher formal–informal wage differential and lower average labor productivity. Increasing enforcement levels of current institutions is also an extremely effective policy to reduce informality, but it has strong adverse effects in most labor market indicators and on welfare. In contrast, reducing taxes or increasing the unemployment benefit have little impact on employment composition (informality remains nearly unchanged), affecting mainly formal–informal wage gap.

The results suggest that the tradeoff between lower informality and higher unemployment does not exist if one considers policies that reduce the costs of being formal, as opposed to those that simply increase the costs of being informal. Furthermore, the simulations

²⁴ Acemoglu (2001) finds the same qualitative results: One of the author's main findings is the positive effect of the unemployment benefit on labor composition and average productivity. Similarly, Fugazza and Jacques (2004) show that higher unemployment benefits increase workers' participation in the formal sector and, in the case of the Italian economy, a slight unemployment reduction.

show that increases in formal entry costs lead to a concomitant rise in unemployment, informal employment and formal–informal wage gap. Thus, this model offers a rationale for the aggregate behavior observed in the Brazilian labor market during the 1990s and early 2000s. During this period, fixed costs of hiring formally increased substantially in Brazil,²⁵ and so did the size of the informal sector, the controlled wage gap and unemployment rate.

Overall, the simulations indicate that the best approach to reduce informality is to adopt policies that diminish the cost of being formal and generate the correct incentives for both workers and firms to enter the formal sector. On the contrary, policies that simply increase the cost of being informal like intensifying auditing or increasing penalties may be effective to reduce informality, but also have substantial negative effects on labor market performance and welfare. Moreover, a substantial part of entry costs is made of bureaucratic costs, which can be removed without any significant cost in terms of foregone government revenues. On the contrary, increasing enforcement demands substantial resources to be employed in auditing and inspecting economic activities all over the country. Considering that reducing entry costs has a positive welfare impact, while increasing enforcement has a negative impact, the results indicate that the former is the most cost effective policy to reduce the size of the informal sector.

Appendix A

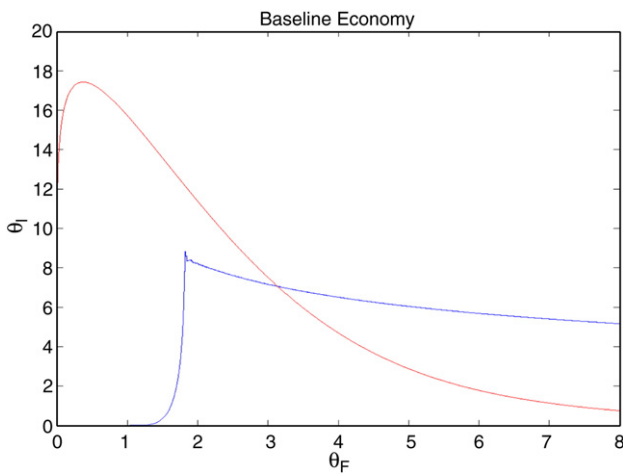


Fig. 4. Equilibrium loci – baseline economy.

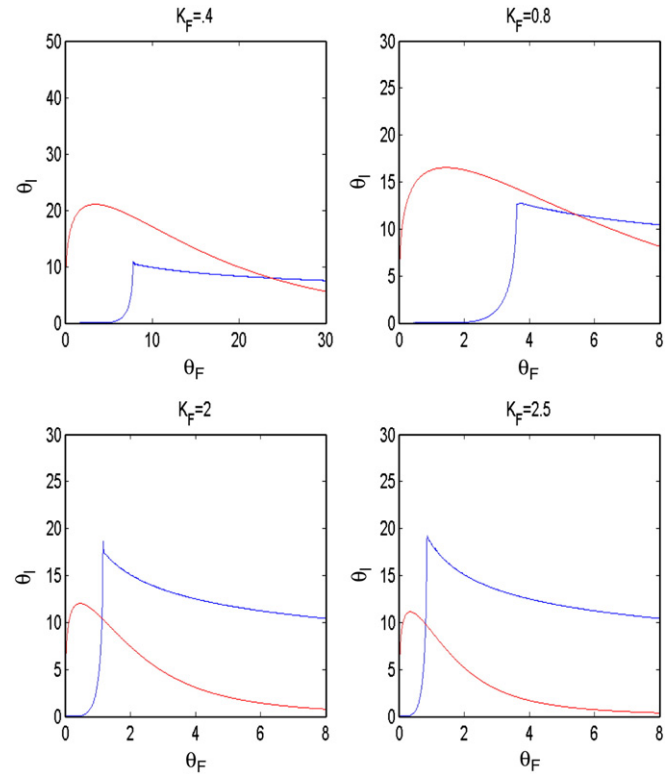


Fig. 5. Equilibrium loci – formal sector entry costs experiments (K_F).

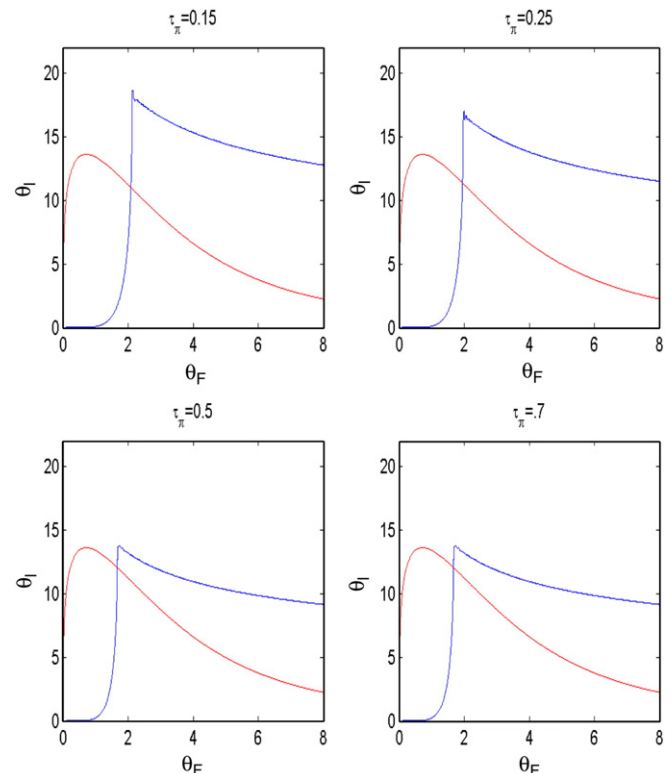


Fig. 6. Equilibrium loci – payroll tax experiments (τ_π).

²⁵ The 1988 constitutional revision significantly increased the fixed costs of hiring formally, as shown in [Gonzaga \(2003\)](#).

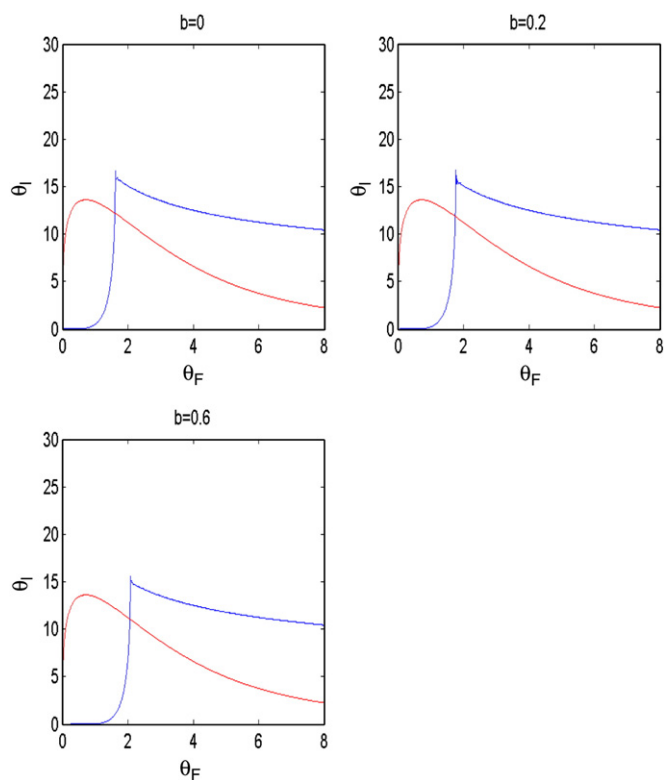
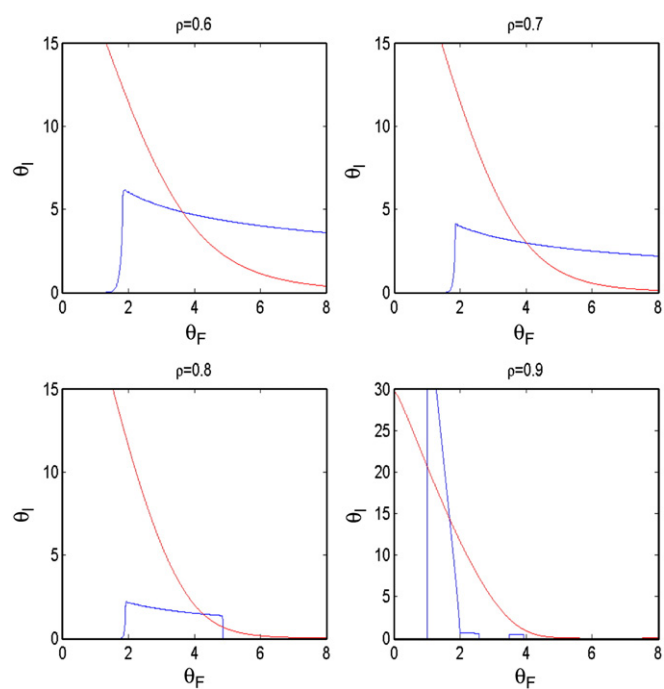
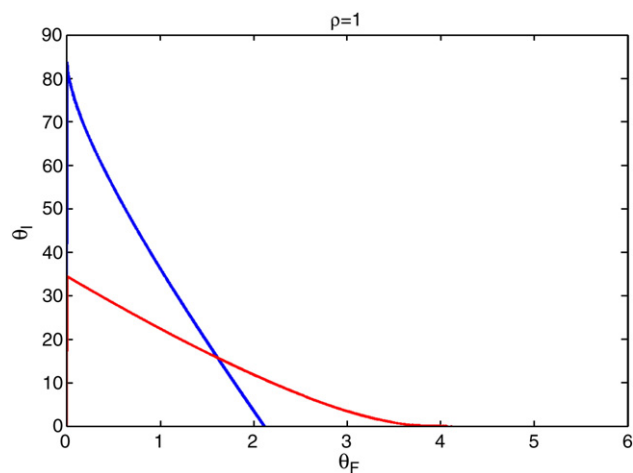
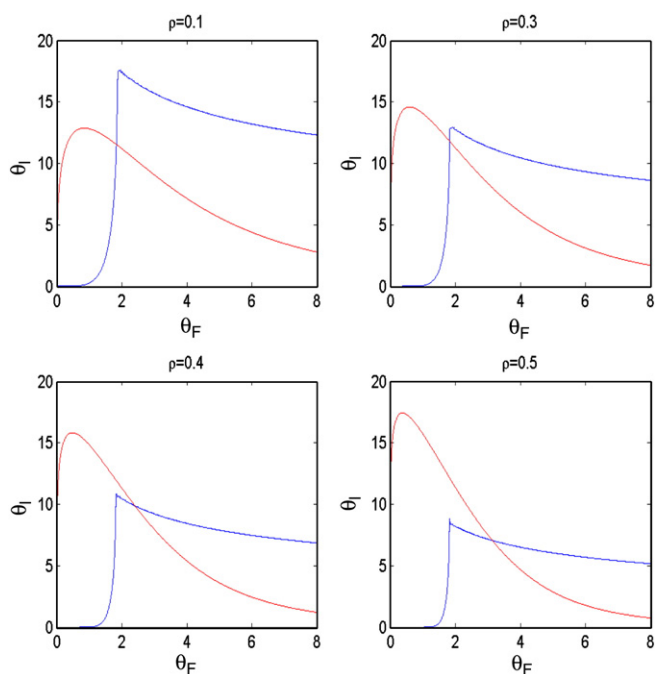
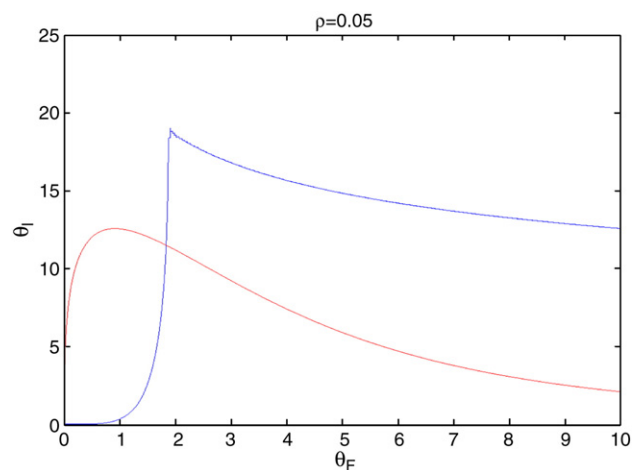


Fig. 7. Equilibrium loci – unemp. benefits experiments (b).

Fig. 9. Equilibrium loci: $0.6 \leq \rho \leq 0.9$ (b).Fig. 10. Equilibrium loci: $\rho = 1$.Fig. 8. Equilibrium loci: $0.1 \leq \rho \leq 0.5$.Fig. 11. Equilibrium loci: $\rho = 0.05$.

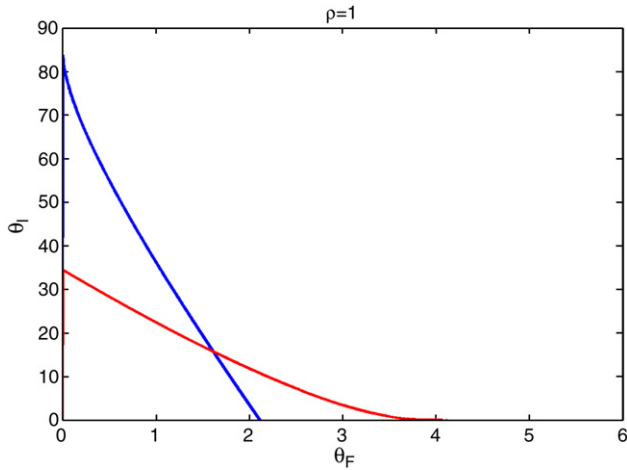


Fig. 12. Equilibrium loci: no institutional differences.

References

- Acemoglu, D., 2001. Good jobs versus bad jobs. *Journal of Labor Economics* 19 (1), 1–22 (January).
- Albrecht, J., L. Navarro, and S. Vroman (2009). The effects of labor market policies in an economy with an informal sector. *Economic Journal*. Forthcoming.
- Auriol, E., Warlters, M., 2005. Taxation base in developing countries. *Journal of Public Economics* 89 (4), 625–646 April.
- Barros, R.P., Corseuil, C.H., 2001. The impact of regulations on Brazilian labor market performance. Inter-American Development Bank Research Paper, No.427.
- Boeri, T., Garibaldi, P., 2005. Shadow sorting. In: Pissarides, C., Frenkel, J. (Eds.), *NBER Macroeconomics Annual*. MIT Press.
- Bosch, M. (2006). Job creation and job destruction in the presence of informal labor markets. Unpublished manuscript.
- Bosch, M., Goni, E., Maloney, W., 2007. The determinants of rising informality in Brazil: evidence from gross worker flows. IZA Discussion Paper, No. 2970.
- Botero, J., Djankov, S., Porta, R., Lopez-De-Silanes, F.C., 2004. The regulation of labor. *The Quarterly Journal of Economics* 119 (4), 1339–1382 November.
- Camargo, J.M., 2002. Fake contracts: justice and labor contracts in Brazil. Brazil Jobs Report, Volume II. World Bank.
- Carneiro, F.G., Henley, A., 2001. Modelling formal versus informal employment and earnings: microeconomic evidence for Brazil. Proceedings of the XXIX Meeting of the Brazilian Economic Association.
- Djankov, S., Porta, R.L., Lopez-De-Silanes, F., Shleifer, A., 2002. The regulation of entry. *The Quarterly Journal of Economics* 117 (1), 1–37 February.
- Fernandes, R., Narita, R., Gremaud, A.P., 2005. Tax reform and the formalization of the economy: simulating different alternatives for Brazil. Paper presented at the X Annual Meeting of the Latin American and Caribbean Economic Association (LACEA).
- Fugazza, M., Jacques, J.-F., 2004. Labor market institutions, taxation and the underground economy. *Journal of Public Economics* 88 (1–2), 395–418 January.
- Goldberg, P.K., Pavcnik, N., 2003. The response of the informal sector to trade liberalization. *Journal of Development Economics* 72, 463–496 December.
- Gonzaga, G., 2003. Labor turnover and labor legislation in Brazil. *Economia* 4 (1), 165–222.
- Heckman, J.J., Pagés, C., 2000. The cost of job security regulation: evidence from Latin American labor markets. NBER Working Paper No.7773.
- Kugler, A., Kugler, M., 2009. Labor market effects of payroll taxes in developing countries: evidence from Colombia. *Economic Development and Cultural Change* 90, 335–358 January.
- Maloney, W., 1999. Does informality imply segmentation in urban labor markets? Evidence from sectoral transitions in Mexico. *World Bank Economic Review* 13 (2), 275–302.
- Pissarides, C., 2000. *Equilibrium unemployment theory*. Basil Blackwell, Oxford.
- Schneider, F., Enste, D., 2000. Shadow economies: size, causes and consequences. *Journal of Economic Literature* 38 (1), 77–114 March.
- Schneider, F., Klinglmaier, R., 2004. Shadow economies around the world: what do we know? IZA Discussion Paper. March.
- Zenou, Y., 2008. Job search and mobility in developing countries. theory and policy implications. *Journal of Development Economics* 86, 336–355.